

NLP – Cosine Similarity Workshop

Objective

To practice using Cosine Similarity as a tool for performing searching for relevant documents based on a given query string

Exercise

You are given a text file called **quotes.txt** that contains 51 quotes. Given a query string, list out the most relevant quotes.

The following is an approach you can consider:

- First, treat each quote as a separate document and compute TFIDF values for each word in the 51 documents (remember to treat each line of quote as a separate document; your corpus is the 51 documents).
- Next, treat the query string as a separate document and compute its TFIDF values as well.
- Then, perform Cosine Similarity between the query string (treat it as a document) and the 51 documents.

A cosine similarity **greater than 0** is regarded as a **relevant document** for the query string. **Rank the documents** (i.e. lines of quote) by their cosine similarity scores (descending order - higher cosine similarity first)

Use the query string – “**life wise choices**” to test your results.

Given the above query string, your results should be:

"Life is the sum of our choices.", [score:0.7393218972723493]

"It is our choices that show what we really are, far more than our abilities.", [score:0.35320464597221546]

"You attract into your life that which you are.", [score:0.32095071358388955]

"Challenges are what make life interesting and overcoming them is what makes life meaningful.", [score:0.24935776599022805]

"Let your life be shaped by decisions you made, not by the ones you didn't.", [score:0.15295031439879156]

"You can live your whole life and never know who you are; until you see the world through the eyes of others.", [score:0.1427887948846313]

"Our greatest fear should not be of failure, but of succeeding at things in life that don't really matter.", [score:0.13006308610420023]

"Don't be afraid your life will end; be afraid that it will never begin.", [score:0.1290320332985028]

"There are two things to aim at in life: first, to get what you want; and, after that, to enjoy it. Only the wisest of mankind achieve the second.", [score:0.11289841969902241]

"To see the world, things dangerous to come to. To see behind walls, to draw closer. To find each other and to feel. That, is the purpose of life.", [score:0.10171860873681889]